Habtamu Asefa

AI Engineer | Model Evaluation and Inference

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GitHub: github.com/habasefa | Portfolio | LinkedIn | Open to remote work and relocation

Python and PyTorch engineer with four years of software experience and hands-on speech-model adaptation and evaluation. Fine-tuning Pocket TTS for Amharic on 200 hours of speech; brings production application delivery, agent integration and technical leadership.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Rust

ML and audio: PyTorch, torchaudio, Hugging Face Transformers, Pocket TTS, VITS

Evaluation: Word error rate, native-speaker evaluation, internal benchmarks

Applications and agents: FastAPI, Node.js/NestJS, Mastra, Agno, React Native

Data and deployment: PostgreSQL, SQLAlchemy, SQLite, Qdrant, Linux, Docker, AWS EC2/S3, GitHub Actions

Experience

Liqawnt Labs | AI Research Engineer | 2026–present

- Fine-tuning Pocket TTS on 200 hours of Amharic speech; preparing audio and text for model adaptation and evaluation.
- Built an internal Amharic benchmark using word error rate and native-speaker ratings; ran reinforcement learning experiments targeting intonation, prosody and accent fidelity.
- Implemented VITS encoder-side fine-tuning in PyTorch with monotonic alignment search and a frozen waveform decoder.

PrepX | Lead Software Engineer | 2024–present

- Built a retrieval-augmented tutoring agent with NestJS and Mastra, using Qdrant and FastEmbed to retrieve textbook and past-exam material with source references.
- Implemented an Agno content-generation workflow spanning research, notes, quizzes, worksheets and flashcards, with explicit context passing between agents.
- Built and launched the React Native/Expo app; students answer approximately 30,000 questions per week, with approximately 200 students joining weekly.
- Integrated Typesense search across 250,000+ items, including questions, curriculum topics and schools; implemented offline SQLite storage and streamed AI responses.
- Contributed to FastAPI/PostgreSQL backend services and the Next.js dashboard.
- Led a four-person engineering team across product development and delivery.

Software Engineer | Client Projects | 2022–present

- Dymnd: worked on real-time habit tracking and an AI habit-development advisor. Themis Bar: fixed mobile bugs, upgraded dependencies and delivered feature updates.
- Contributed to Remimb's React Native Bluetooth-keyring companion app and implemented React Native/Expo marketplace screens for browsing, cart, checkout and orders.

Selected Projects

react-native-ajora | Author and Maintainer. Published a React Native AI chat library used in PrepX, supporting streamed agent responses, tool-call rendering and client-side tool execution. Source

GoNumbers | Telecom Application. Developed React Native calling features with SIP.js, FreeSWITCH, WebRTC audio and native call controls.

Education

Addis Ababa University / AAiT | BSc in Biomedical Engineering | 2024